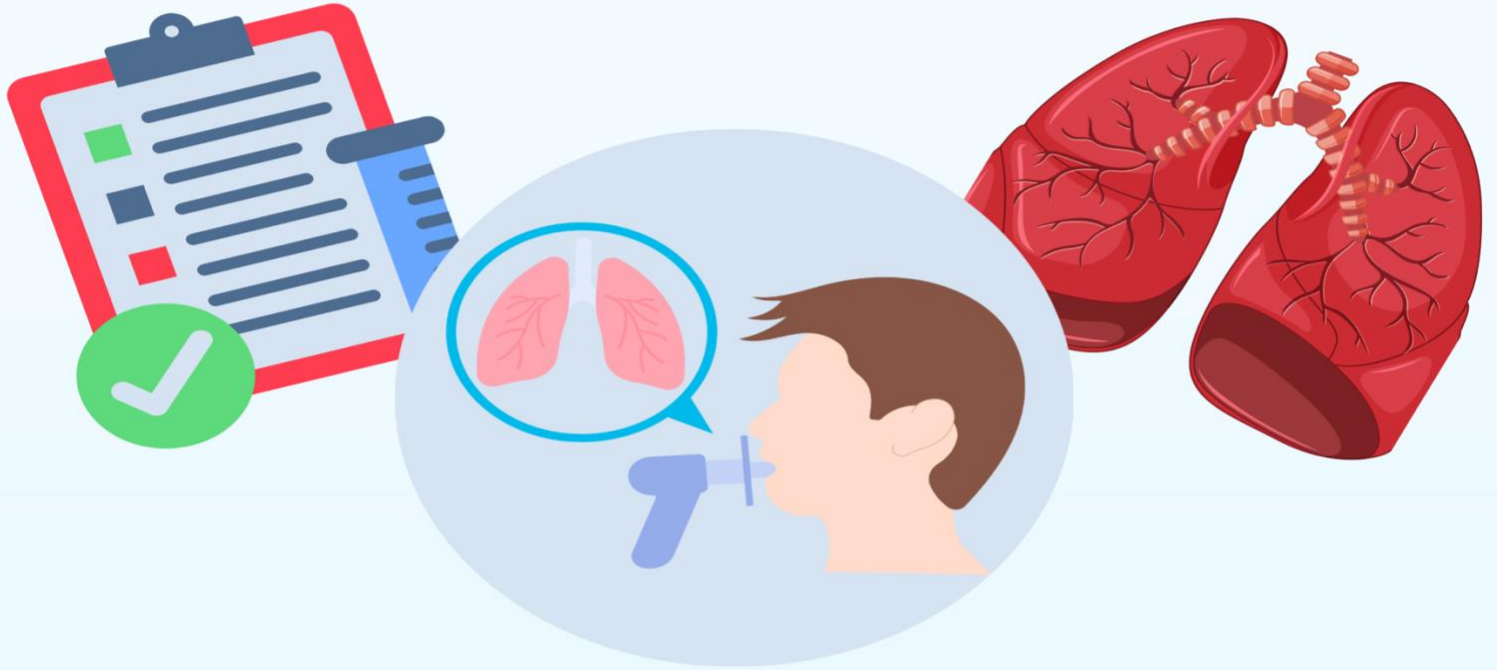




GUIDANCE PROTOCOL FOR SPIROMETRY



برنامج التصدي للأمراض المزمنة غير المعدية
في الرعاية الصحية الأولية
NCD program in PHC

Under the supervision of

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Disclaimer

This clinical guidance is based on published data and evidenced based medicine existing at the production of this work in November 2024. The authors are not responsible in case of subsequent updates in the literature and changes in medical practices regarding topics pertaining to or areas covered in these guidelines.

Introduction:

- A spirometer is a device for measuring timed expired and inspired volumes; it indicates how quickly and effectively the lungs can be emptied and filled.
- It is a reliable method of differentiating between obstructive airways disorders (e.g. chronic obstructive pulmonary disease, asthma) and restrictive diseases (e.g. fibrotic lung disease).
- It can also help to monitor disease severity.
- Spirometry is the gold standard method for the diagnosis, assessment and monitoring of COPD and can also be used to demonstrate obstruction of airways and reversibility in the diagnosis of asthma.

Referral policy:

- Age 5 years and above
- Referral should be from non-communicable disease (NCD) clinics through a written referral form.

Indications of spirometry:

- Evaluation of symptoms suspecting asthma.
- patient with asthma: at diagnosis or start of treatment, after 3-6 months of ICS-containing treatment to assess best FEV1, and periodically thereafter (at least every 1-2 years, more frequently in higher risk patients including those with exacerbations and those at risk of decline in lung function)
- Evaluation of uncertainty of diagnosis

Precaution related to respiratory viral infection:

- PFT equipment should be fitted with single-use mouthpieces and disposable in-line bacterial and viral filters
- Recognizing that the time needed for optimal clearance of aerosol particles varies among laboratories, optimal closure between patients may vary from 20 minutes to 3 hours (depending on ventilation of the particular space)

Preparation - pre procedure:

1. Minimal timing of Holding bronchodilator medications:

BRONCHODILATOR	WITHHOLDING TIME
Short – acting beta 2 agonist (SABA) Ventolin	4 hours
Short – acting muscarinic antagonists (SAMAs - Atrovent) / inhaled corticosteroids	12 hours
Long – acting beta 2 agonists (LABA)	24 hours
Long – acting muscarinic antagonists (LAMAs)	36 hours

- Abstain from smoking for at least 6 hours prior to testing.
- Avoid heavy and dense meals at least 2 hours prior to testing.

2. Procedure instructions to the patient:

1. Place the mouthpiece inside the mouth
2. Inhale and exhale normally around 3 times.
3. Inhale quickly and as fast as possible until lungs are filled.
4. Exhale forcefully as fast and as hard as possible.
 - Patient may feel dizzy and needs to rest between the attempts
 - the procedure lasts 15 minutes
 - To test for bronchodilator responsiveness, SABA by MDI will be administered (4 puffs), then Spirometry will be repeated 10-15 minutes after that.

Interpretation

	FEV1	FVC	FEV1/FVC
NORMAL	>70%	>70%	>70%
OBSTRUCTIVE	<70%	>70%	<70%
RESTRICTIVE	<70%	<70%	>70%

SEVERITY	FEV1/FVC
MILD	>80% PREDICTED
MODERATE	50-79% PREDICTED
SEVERE	30-49% PREDICTED
VERY SEVERE	<30% PREDICTED

Reversibility test:

- It can be useful to assess reversibility with a bronchodilator if considering asthma as a cause of obstructive airway disease.
- Administer 400 micrograms of salbutamol (4 puffs Ventolin inhaler, or 1 ml Ventolin nebulizer) and repeat spirometry after 15 minutes
- A 12% and minimum of 200 ml increase in FVC and/or FEV1 is considered significant
- The presence of reversibility is suggestive of a diagnosis of asthma.

Normal value:

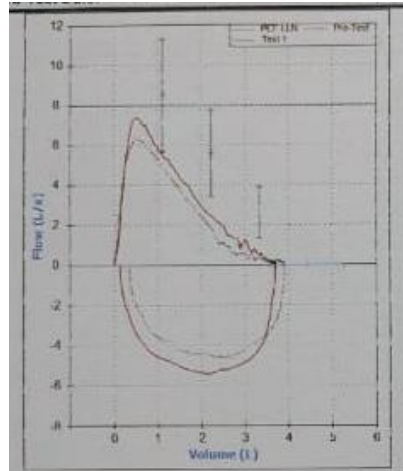
- A normal spirometry test is characterized by FVC, FEV1, and FEV1/FVC all within the normal range.
- values in the normal range cannot completely rule out lung disease especially in asthma

Restrictive ventilatory pattern

- Due to conditions where lung volume is reduced - eg, fibrosing alveolitis, scoliosis.
- The FVC and FEV1 are reduced proportionately:
- FVC reduced

Obstructive ventilatory pattern

- Due to conditions in which airways are obstructed due to diffuse airways narrowing of any cause - eg, asthma, COPD, extensive bronchiectasis, cystic fibrosis., emphysema
- The FVC and FEV1 are reduced disproportionately:
- FEV1 reduced <80%
- FVC normal or reduced (but to a lesser extent than FEV1)
- FEV1/FVC reduced <70%



Parameter	Pred	11N	Pre Best	% Pre Pred	Pre Z-Score	Post Best	%Post Pred.	Post Z-Score	% Pre Post
FVC (L)	4.43	3.58	3.91	88	-1.00	3.71	84	-1.39	-5.1
FEV1 (L)	3.74	3.03	2.69*	72	-2.43	3.02*	81	-1.66	12.3
FEV1 Ratio	0.83	0.71	0.69*	83	-1.91	0.81	98	-0.22	12.0
FEV6 (L)	4.43	3.58	3.89	88	-1.04	3.71	84	-1.39	-4.6
PEF (L/min)	596	478	380*	64	-3.00	443*	74	-2.13	16.3
FEF25-75 (L/s)	5.06	3.35	1.53*	30	-3.39	2.84*	56	-2.13	85.6

Steps of reading a Spirometry:

- Review patient demographics
- Review the clinical history (consider the clinical context)
- Assess the quality of the result
- Review the technician comments
- The graphs should also be smooth and free from abnormalities caused by coughing during expiration, extra breath during expiration, slow start to forced expiration or sub-maximal effort
- Interpret the spirometry findings (FVC, FEV1, their ratios, pre and post bronchodilator)